Weekly Research Report — Week 3

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Summary of Work Done

Research questions were reformulated as hypotheses per supervisor direction, the paper skeleton was built in the IEEE conference template, and the first working retrieval pipeline was implemented and evaluated. The initial result was counter-intuitive and shaped the framing of the Introduction.

Hypotheses

The four open research questions from Week 2 were restated as testable hypotheses, with the hybridisation target incorporated as directed:

- **H1 Comparative efficacy** — a RAG framework integrated with a dynamic medical ontology yields statistically significant improvements in structural precision and clinical accuracy over zero-shot baselines
- **H2 Uncertainty awareness** — augmenting the LLM with a probabilistic uncertainty-quantification layer increases the proportion of uncertainty-aware outputs without compromising factual correctness
- **H3 Temporal relevance** — outputs augmented by real-time evidence retrieval and structured chain-of-thought achieve greater diagnostic accuracy and better-calibrated confidence than unaugmented LLMs

First experimental pipeline

Environment established in Google Colab. Backbone model: `llama-3.3-70b-versatile` accessed via the Groq API, chosen to avoid consuming the limited OpenAI budget during pipeline development.

Two architectures implemented and evaluated on a MedQA-USMLE sample:

Architecture	Accuracy
Baseline LLM, no retrieval	70%
Flat-vector RAG (FAISS, all-MiniLM-L6-v2)	65%

Retrieval degraded performance by 5 percentage points. This was not the expected direction. The likely explanation is retrieval noise: passages scoring highly on embedding similarity are topically related to the clinical question but do not contain the specific causal reasoning required to answer it, and the injected context displaces the model's own correct reasoning.

This observation became the motivating problem for the Introduction — that standard RAG can actively harm clinical QA — and the argument for structurally precise retrieval as the alternative.

Writing

Abstract and Introduction drafted in IEEE two-column format. The Introduction opens with the parametric knowledge bottleneck, develops the solution space across flat-vector RAG and its precision failure, and positions knowledge-graph augmentation as the structurally precise alternative, covering Local, Global and DRIFT strategies.

Supervisor direction received

Draft to be submitted as MS Word rather than PDF, with the Colab notebook accessible. An IEEE conference submission raised as the near-term target. Pace flagged as a concern.

Plan for Week 4

1. Implement the GraphRAG pipeline — entity and relation extraction, graph construction, subgraph retrieval
2. Evaluate GraphRAG against the two existing architectures
3. Draft the Methodology section
4. Share the Word draft and notebook

Blockers

None outstanding. Groq API access is sufficient for development; the decision on paid API usage can be deferred until the evaluation scales.